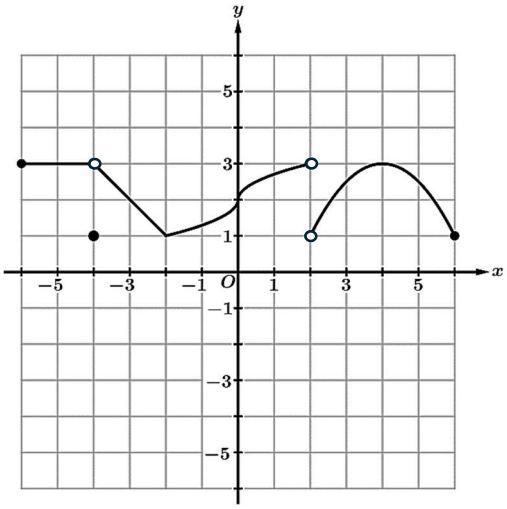
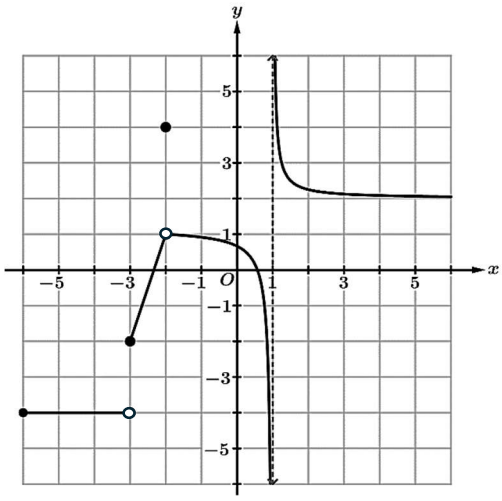
Worksheet B Review of Unit 1
AP Calculus

Name Answers

Directions: Use the graphs to find each limit, if it exists.



Graph of f



Graph of g

1. $\lim_{x\to 2^-} f(x) = 3$
2. $\lim_{x\to 2^+} f(x) = 1$
3. $\lim_{x\to 1^-} g(x) = -\infty$
4. $\lim_{x\to 1^+} g(x) = +\infty$
5. $\lim_{x\to 2} f(x) =$ DNE
6. $f(2)$ undef.
7. $\lim_{x\to -2} g(x) = 1$
8. $g(-2) = 4$
9. $\lim_{x\to -4} f(x) = 3$
10. $f(-4) = 1$
11. $\lim_{x\to -3} g(x) =$ DNE
12. $\lim_{x\to \infty} g(x) = 2$

13. Find and classify any discontinuities for the graph of f.

$x=-4$ removable $x=2$ jump

14. For which value(s) of x does the graph of g have a nonremovable discontinuity?

$x=-3$ jump $x=1$ infinite

15. For which value(s) of c does $\lim_{x\to c} f(x) = 2$?

$x=-3$, $x=0$, $x=2.5$, $x=5.5$

Directions: Find and classify any discontinuities for the following functions.

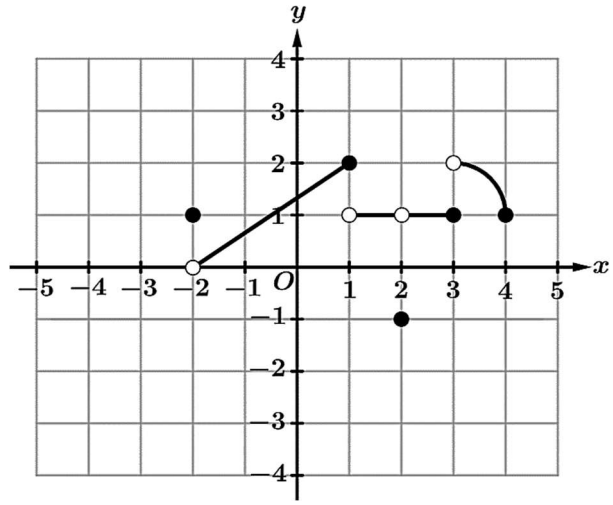
16. $k(x) = \frac{x^2+8x+1}{x^2-3x-1}$

$x=-3$ removable (hole)
$x=6$ infinite

17. $w(x) = \begin{cases} 8x+3, & \text{if } x \le -1 \\ x^2-4, & \text{if } x > -1 \end{cases}$

$x=-1$ jump

The graph of h is shown. Use the graph to answer the following true or false questions.



Graph of h

18. $\lim_{x\to 2} h(x) = -1$ F

19. $\lim_{x\to 1^+} h(x) = 1$ T

20. $\lim_{x\to -2^+} h(x) = 1$ F

21. $\lim_{x\to 2} h(x)$ exists T

22. $\lim_{x\to 3} h(x) = 1$ F

23. $\lim_{x\to 1} h(x)$ DNE T

24. $\lim_{x\to 3^-} h(x) = 2$ F

25. $\lim_{x\to 0^-} h(x) = \lim_{x\to 0^+} h(x)$ T

26. $\lim_{x\to 0} h(x)$ exists T

27. For which value(s) of x does the graph of h have a jump discontinuity?

$x=1$ and $x=3$

Directions: For the following piecewise functions, find the value of k that makes the function continuous, or state that no such value exists.

28. $f(x) = \begin{cases} \frac{9x^2-4}{3x+2}, & \text{if } x \neq -\frac{2}{3} \\ k, & \text{if } x = \frac{2}{3} \end{cases}$

$\boxed{k=0}$

29. $g(x) = \begin{cases} x^2 - 2, & \text{if } x < 3 \\ kx + 4, & \text{if } x \geq 3 \end{cases}$

$\boxed{k=1}$

30. $m(x) = \begin{cases} 3x - 1, & \text{if } x < 2 \\ \ln(k), & \text{if } x \geq 2 \end{cases}$

$\boxed{k=e^5}$

31. $p(x) = \begin{cases} \frac{x^2-x-6}{x^2-4}, & \text{if } x \leq -2 \\ kx^2, & \text{if } x > -2 \end{cases}$

$p(-2)$ = undefined

There is no value of k that will make $p(x)$ continuous

Directions: Find the following limits or state that the limit does not exist.

32. $\lim_{x\to\infty} \frac{4x-2}{5-8x}$ $\boxed{-\frac{1}{2}}$

33. $\lim_{x\to-\infty} \frac{4x-2}{5-8x}$ $\boxed{-\frac{1}{2}}$

34. $\lim_{x\to -4^+} \frac{x+1}{x+4}$ $= \boxed{-\infty}$

Directions: Find the following limits or state that the limit does not exist.

35. $\lim_{x\to -6} \frac{x-6}{x^2+36}$

$\boxed{-\frac{1}{6}}$

36. $\lim_{x\to 6} \frac{x-6}{x^2-4x-12}$

$\boxed{\frac{1}{8}}$

37. $\lim_{x\to 5} \frac{2x^2-5x-2}{x-5}$

$\boxed{15}$

38. $\lim_{x\to \infty} \frac{(4x-1)(5-3x)}{(3x+2)^2}$

$\boxed{-\frac{4}{3}}$

39. $\lim_{x\to \infty} \frac{2^x+3x}{x^{99}-2^x}$

$\boxed{-1}$

40. $\lim_{x\to -\infty} \frac{4^x-5}{2-e^x}$

$\boxed{-\frac{5}{2}}$

41. $\lim_{x\to 8} \frac{\sqrt{x+1}-3}{x-8}$

$\boxed{\frac{1}{6}}$

42. $\lim_{x\to 3^-} \frac{|x-3|}{x-3}$

$\boxed{-1}$

43. $\lim_{x\to \infty} \frac{4-7x}{\sqrt{9x^2+2x-7}}$

$\boxed{-\frac{7}{3}}$

44. $\lim_{x\to -1} \frac{4|x+1|}{x+1}$ $= \boxed{\text{DNE}}$

45. $\lim_{x\to 5^-} \frac{2x-10}{8|x-5|}$

$\boxed{-\frac{1}{4}}$

46. $\lim_{x\to -\infty} \frac{\sqrt{4x^2+7}}{5x-1}$

$\boxed{-\frac{2}{5}}$

47. $\lim_{x\to 0} \frac{3x^2-5x^4}{2x^2+4x^6}$

$\boxed{\frac{3}{2}}$

48. $\lim_{x\to \infty} \frac{\ln(x)}{e^x}$

$= \boxed{0}$

49. $\lim_{x\to -3} \frac{x^2-6x-27}{x+3}$

$\boxed{-12}$

$$f(x) = \begin{cases} 3x - x^2, & \text{if } x < 4 \\ 2x - 12, & \text{if } x \geq 4 \end{cases}$$

50. The piecewise defined function f is given above. Show that f is continuous at $x = 4$. Justify your answer.

$f(4) = -4$

$\lim_{x \to 4} f(x) = -4$

$\lim_{x \to 4^-} f(x) = \lim_{x \to 4^-} (3x - x^2) = -4$

$\lim_{x \to 4^+} f(x) = \lim_{x \to 4^+} (2x - 12) = -4$

$f(x)$ is continuous at $x=4$ because $\lim_{x \to 4} f(x) = f(4)$

$$g(x) = \begin{cases} \ln(x^2 + 1), & \text{if } x < 0 \\ \cos(\pi x), & \text{if } x \geq 0 \end{cases}$$

51. The piecewise defined function g is given above. Is g continuous at $x = 0$? Give a reason for your answer.

$g(0) = \cos(0) = 1$

$\lim_{x \to 0} g(x) = \text{DNE}$

$\lim_{x \to 0^-} g(x) = \lim_{x \to 0^-} (\ln(x^2+1)) = \ln 1 = 0$

$\lim_{x \to 0^+} g(x) = \lim_{x \to 0^+} (\cos(\pi x)) = \cos 0 = 1$

g is discontinuous at $x=0$ because $\lim_{x \to 0} g(x)$ DNE

t	1	4	5	7	12
$a(t)$	-2	0	-3	5	-5

52. For $1 \leq t \leq 12$, a particle accelerates back and forth along a horizontal axis. The acceleration of the particle is modeled by the continuous function $a(t)$. Selected values for $a(t)$ are shown in the figure above. Must there be a time c, $1 < c < 12$, such that $a(c) = -4$? Justify your answer.

Since $a(t)$ is continuous on the closed interval $[1,12]$, it is continuous on the closed interval $[7, 12]$, where $a(7) \neq a(12)$ and -4 is a number between $a(7) = 5$ and $a(12) = -5$. So, the IVT guarantees there is at least one value c in $(7, 12)$, which is in the interval $(1, 12)$, such that $a(c) = -4$.